HARSH GAURAV

Software Engineer | Next.js, React, TypeScript, Node.js, Python, PostgreSQL

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PROFESSIONAL SUMMARY

Software engineer with about a year of experience on a production platform for ad creatives and performance reporting (product name under NDA). I work mainly with Next.js, TypeScript, Node.js, Python and PostgreSQL, and I'm comfortable with databases, background jobs and cloud setups on Vercel, AWS and GCP. Recent results: a dashboard refresh that took over a minute now finishes in under 2 seconds, production deploys are over 90% faster, and a new Slack notification service runs on GCP Pub/Sub. Looking for full-stack or backend roles.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), SQL, Python (background services), HTML5, CSS3

Frontend: Next.js 15 (App Router), React, React Hooks, streaming UI (SSE), responsive design, role-based dashboards

Backend & APIs: Node.js, Express, REST API design, event-driven architecture, microservices, outbox pattern, background jobs (Celery, Redis), GCP Pub/Sub, JWT authentication

Databases: PostgreSQL, BigQuery, Firestore, pgvector, Prisma ORM, Drizzle ORM, schema design, Prisma Migrate, query optimization

Cloud & DevOps: Vercel (serverless/Edge), AWS (EC2, S3, CloudWatch), GCP (Cloud Run, Pub/Sub, Firestore, GCS, BigQuery), Docker, GitHub Actions CI/CD, distributed locks

AI IDEs & Agents: Cursor, Claude Code, Google Antigravity, MCP (Model Context Protocol), AI coding-agent workflows

Practices: Code review, feature-branch workflow, root-cause analysis, Vitest, CI schema guards, PRD authorship

WORK EXPERIENCE

Software Engineer | Code Garage Tech **Nov 2025 – Present**

Project: enterprise ad-creative & performance platform (product name under NDA) | *Next.js 15, TypeScript, Node.js, Prisma/PostgreSQL, Python/Celery, BigQuery, Vercel*

- Designed and built the workflow engine for video production: 5 production stages with rules that stop anyone skipping a step, using database triggers and an outbox pattern. Also built the "My Tasks" dashboards for editors and strategists, a Workflow Inspector screen, and moved 14k+ existing concepts over with zero downtime.
- Built a notification service that posts to Slack through GCP Pub/Sub, across 3 repositories: Next.js event producers (idempotent, 6 typed events), a Python consumer on Cloud Run, Firestore to track delivery, and retries with exponential backoff from 10 seconds up to 600 seconds.
- Brought an analytics dashboard refresh from 60+ seconds down to under 2 seconds. The old setup ran redundant BigQuery cache-warming scans; I replaced them with a hybrid read that uses cached PostgreSQL history plus a live BigQuery query for just yesterday and today.
- Found why production deploys took about 43 minutes: a 2.3 GB virtual environment was going into the Docker build. Excluded it with .dockerignore and merged duplicate service builds, which cut deploy time by over 90%.
- Built a Dropbox video ingestion pipeline in Python and Celery. It uses content fingerprinting (rev/serverModified) to spot edited or re-uploaded files, starts AI compliance checks in the background, and safely cleans up orphaned media in GCS.
- Tracked down a bug that only happened in production: Vercel's Edge CDN was stripping empty query parameters. Added client-side short-circuit guards in the React data-fetching hooks, which removed 100% of the wasted BigQuery and API calls for empty filters and reduced cloud costs.
- Moved the unit-economics (contribution margin) recompute from PostgreSQL stored procedures to background Python/Celery workers. Added Redis distributed locks, 5-second status polling and a safety guard for unknown status, so two recomputes can't overlap and nobody edits mid-calculation.
- Tested and set up an MCP (Model Context Protocol) server that maps the whole repository into a local database, so AI coding agents reuse the map instead of searching the code again. I use it in my own agent workflow.

Full-Stack Developer Intern | GeeksforGeeks **Jul 2025 – Oct 2025**

- Built full-stack apps with React, Node.js, Express and PostgreSQL using MVC; added JWT authentication, REST APIs and CRUD operations.
- Added client- and server-side validation with clear error handling, so invalid data is blocked at both ends.
- Wrote unit tests with Jest and React Testing Library, and improved rendering with React.memo, lazy loading and code splitting.

PROJECTS

DocChat: Document Q&A with Cited Answers (RAG) | Next.js, TypeScript, PostgreSQL + pgvector, Drizzle ORM, Google Gemini, Vercel
Live: docchat-kohl.vercel.app | Code: github.com/hgroy/docchat

- Built a RAG app that answers questions about uploaded PDF, TXT and Markdown files and cites the page each answer came from. Each page is split into chunks, embedded with Gemini and searched by cosine similarity in pgvector, with versioned SQL migrations.
- Answers stream in token by token (NDJSON). Every query is limited to the signed-in user. Deployed on Vercel with Neon serverless PostgreSQL.

Blog Publishing Platform | React, Node.js, Express, PostgreSQL, JWT

- Built a full-stack blog with protected routes, JWT authentication, REST APIs (MVC), a normalized PostgreSQL schema and validation on both client and server.

EDUCATION

B.Tech, Information Technology | Arya College of Engineering & IT, Jaipur **Jul 2021 – Jun 2025**

CGPA: 8.9/10